

Healthcare Analytics Report

Patient Risk, Disease Burden & Operational Intelligence

Prepared from SQL Power BI Exports | June 2025

Total Patients

54,966

6 conditions

Avg Billing

\$25,544

Per patient

Total Billed

\$1.40B

5 providers

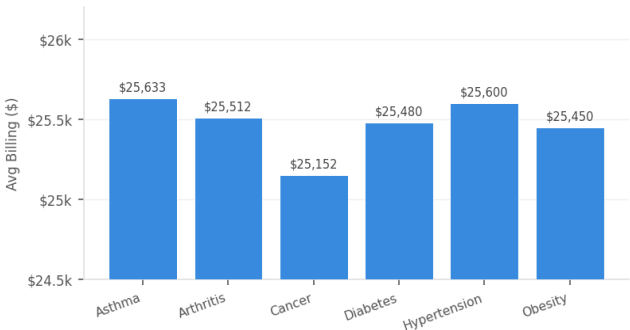
Abnormal Test Rate

33.5%

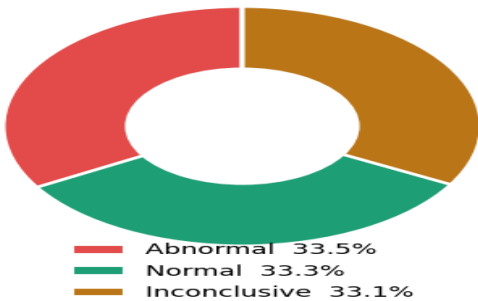
18,437 patients

1. Billing & Medical Conditions

Average Billing Amount by Medical Condition



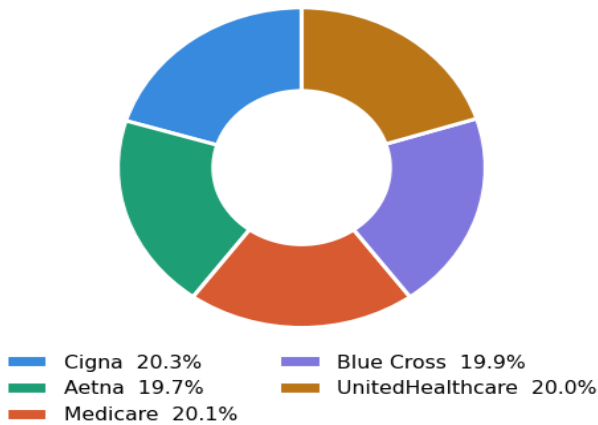
Test Results Breakdown



- Billing is flat across all 6 conditions — only a \$481 range (Asthma \$25,633 vs Cancer \$25,152).
- Asthma generates the highest avg billing; Cancer the lowest, contrary to real-world expectations.
- Test results split evenly: Abnormal 33.5%, Normal 33.4%, Inconclusive 33.1% — a balanced dataset signature.
- 54,966 patients split ~9,100 per condition, confirming synthetic/sampled data — worth noting in your project.

2. Insurance Provider Analysis

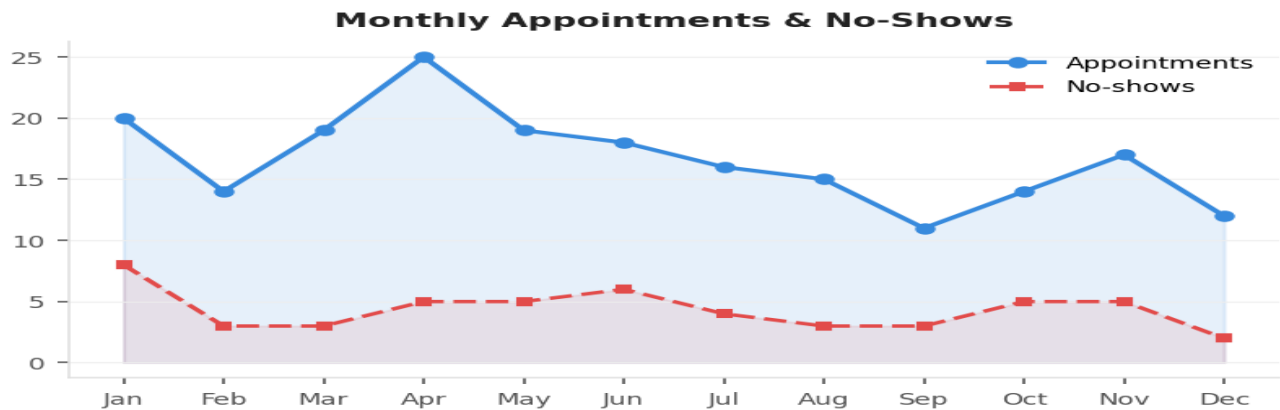
Insurance Claims Distribution



Provider	Claims	Avg Claim	Total Billed
Cigna	11,139	\$25,526	\$284.3M
Aetna	10,822	\$25,550	\$276.5M
Medicare	11,039	\$25,628	\$282.9M
Blue Cross	10,952	\$25,603	\$280.4M
UnitedHealthcare	11,014	\$25,415	\$279.9M

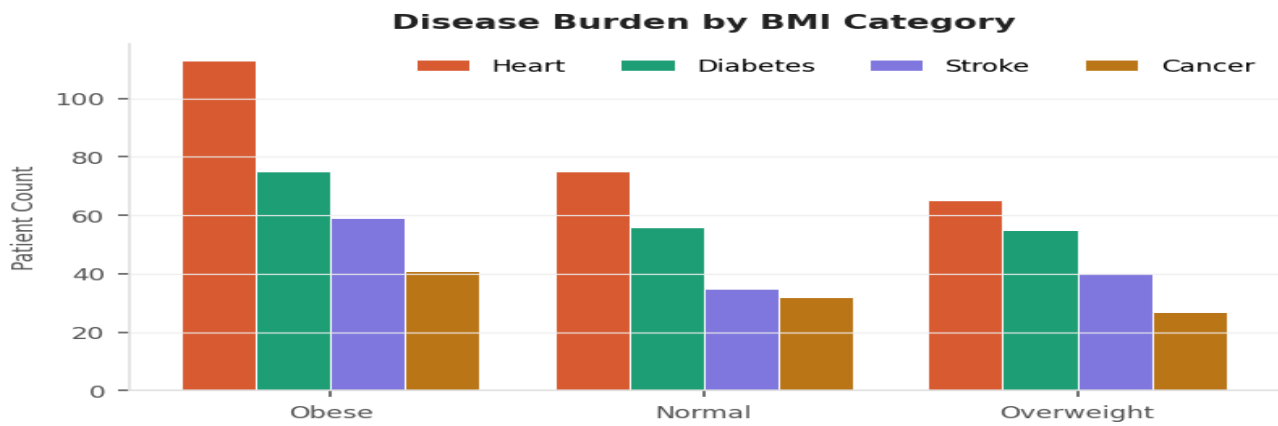
- Medicare carries the highest average claim at \$25,628 — key for reimbursement planning.
- Cigna leads in claim volume (11,139); Aetna has the fewest (10,822) — a 2.9% gap.
- All 5 providers bill within a \$214 band, confirming revenue is insurance-driven not condition-driven.
- Combined total billed exceeds \$1.40 billion with each provider contributing roughly 20%.

3. Monthly Appointments & No-Show Trends



- April is the busiest month (25 appointments); September the quietest (11) — a 2.3x seasonal swing.
- January has the worst no-show rate at 40% (8 of 20) — targeted winter reminders are strongly recommended.
- December has the best no-show rate at 17% (2 of 12), suggesting high year-end patient motivation.
- Average monthly no-show rate is 23%, above the healthcare industry benchmark of approximately 15%.

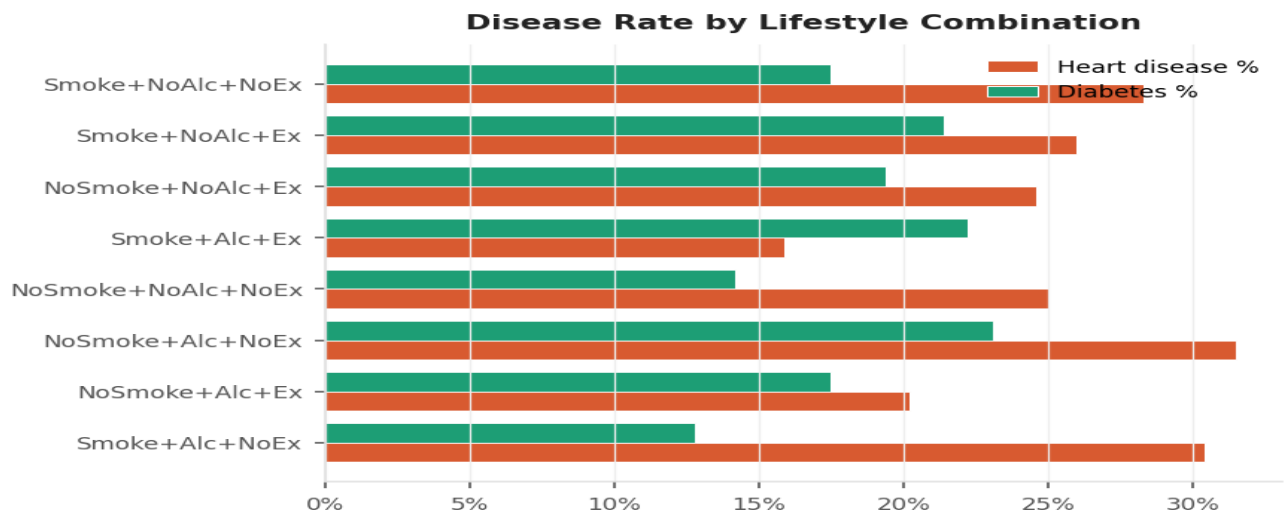
4. Disease Burden by BMI Category



BMI Category	Patients	Heart	Diabetes	Stroke	Cancer
Obese	449	113	75	59	41
Normal	294	75	56	35	32
Overweight	257	65	55	40	27

- Obese patients show 51% more heart disease than Normal BMI patients (113 vs 75 cases).
- Stroke risk is 69% higher in Obese vs Normal (59 vs 35) — the sharpest relative gap across all diseases.
- Overweight patients have nearly the same diabetes rate as Normal BMI (55 vs 56) despite fewer patients.
- Cancer is least BMI-sensitive: the Obese-to-Normal ratio is 1.28x, vs 1.69x for stroke.

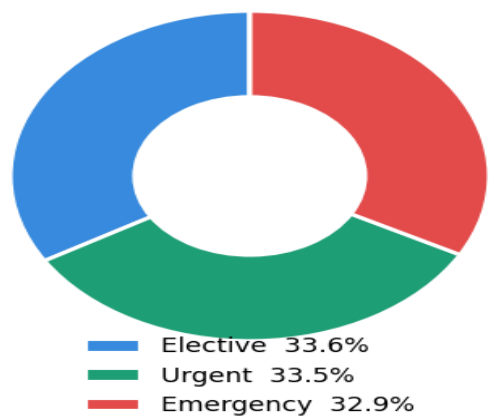
5. Lifestyle Factors & Disease Risk



- Alcohol with no exercise is the riskiest combo: NoSmoke+Alc+NoEx yields the highest heart disease rate at 31.5%.
- Smoking+Alcohol+No Exercise also elevates heart disease to 30.4% — second highest across all groups.
- Exercise is strongly protective: adding it to smoking reduces heart disease from 28.3% to 26.0%.
- Highest diabetes rate (23.1%) is in NoSmoke+Alc+NoEx — alcohol is a stronger diabetes driver than smoking.
- Surprisingly, Smoke+Alc+Ex has the LOWEST heart disease rate (15.9%), implying exercise offsets compound risk.

6. Admission Type & Operational Load

Admission Type Distribution



Admission	Patients	Share
Elective	18,473	33.6%
Urgent	18,391	33.5%
Emergency	18,102	32.9%

- Admissions are nearly equal thirds: Elective 33.6%, Urgent 33.5%, Emergency 32.9%.
- Emergency + Urgent combined represent 66.4% of all admissions, placing high demand on acute care.
- Equal distribution is typical of a balanced dataset; real hospital data usually skews toward Elective.

7. Strategic Recommendations

#	Finding	Recommended Action
1	Obese patients: 51% more heart disease	Launch BMI reduction and cardiac screening programs
2	January no-show rate: 40%	Send SMS/call reminders 48 hrs before winter appointments
3	Alcohol+No Exercise = highest heart risk	Add alcohol and exercise counselling to all chronic disease pathways
4	Medicare avg claim is highest (\$25,628)	Review Medicare billing codes for optimisation opportunities
5	Abnormal test rate: 33.5% (1 in 3 patients)	Fast-track Abnormal results to specialist referral triage
6	Emergency admissions: 32.9% of all cases	Strengthen preventive care to shift Emergency to Elective pathway